

Multiple Choice (20 marks)

1. Which of the following is a common renewable resource found in deserts?
 - (a) biodiesel
 - (b) uranium
 - (c) natural gas
 - (d) solar energy
2. which of these would you most likely find in a log with a hole in it?
 - (a) a fish with large gills
 - (b) a dinosaur from the Jurassic era
 - (c) none of these
 - (d) a small sized raccoon
3. What sport is featured in the 1996 movie 'Kingpin'?
 - (a) wrestling
 - (b) golf
 - (c) chess
 - (d) bowling
4. What tool is used to determine the mass of an object?
 - (a) Balance
 - (b) Meter stick
 - (c) Thermometer
 - (d) Graduated cylinder
5. Which of these quantities decreases as the inverse square of distance for distances far from the objects producing the fields?
 - (a) the electric field produced by a finite-length charged rod
 - (b) the electric field produced by an infinitely long charged cylinder
 - (c) the electric field produced by an infinite plane of charge
 - (d) the magnetic field produced by an infinitely long, straight current-carrying wire
6. In which layer of the Sun does nuclear fusion occur?
 - (a) core

- (b) radiative zone
 - (c) convection zone
 - (d) chromosphere
7. Corn would likely provide nutrients to
- (a) a rock
 - (b) a ice
 - (c) a waterfall
 - (d) a raven
8. The electrolysis of molten magnesium bromide is expected to produce
- (a) magnesium at the anode and bromine at the cathode
 - (b) magnesium at the cathode and bromine at the anode
 - (c) magnesium at the cathode and oxygen at the anode
 - (d) bromine at the anode and hydrogen at the cathode
9. Which of these best defines the term "pollutant"?
- (a) anything humans add to the atmosphere and oceans
 - (b) any substance that is harmful to the environment
 - (c) any substance that is a waste product of industry
 - (d) a substance that does not break down over time
10. War became possible between nation states in the nineteenth century because:
- (a) the world was divided into several rival overseas empires
 - (b) industrialization provided better transport, technology and administration
 - (c) there was intense economic, political and military competition
 - (d) all of the above

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a function to move all the numbers in it to the given string.
2. Write a python function to find the first digit of a given number.

End of Paper